FlashVSR: Towards Real-time Diffusion-Based Streaming Video Super Resolution

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https://zhuang2002.github.io/FlashVSR/

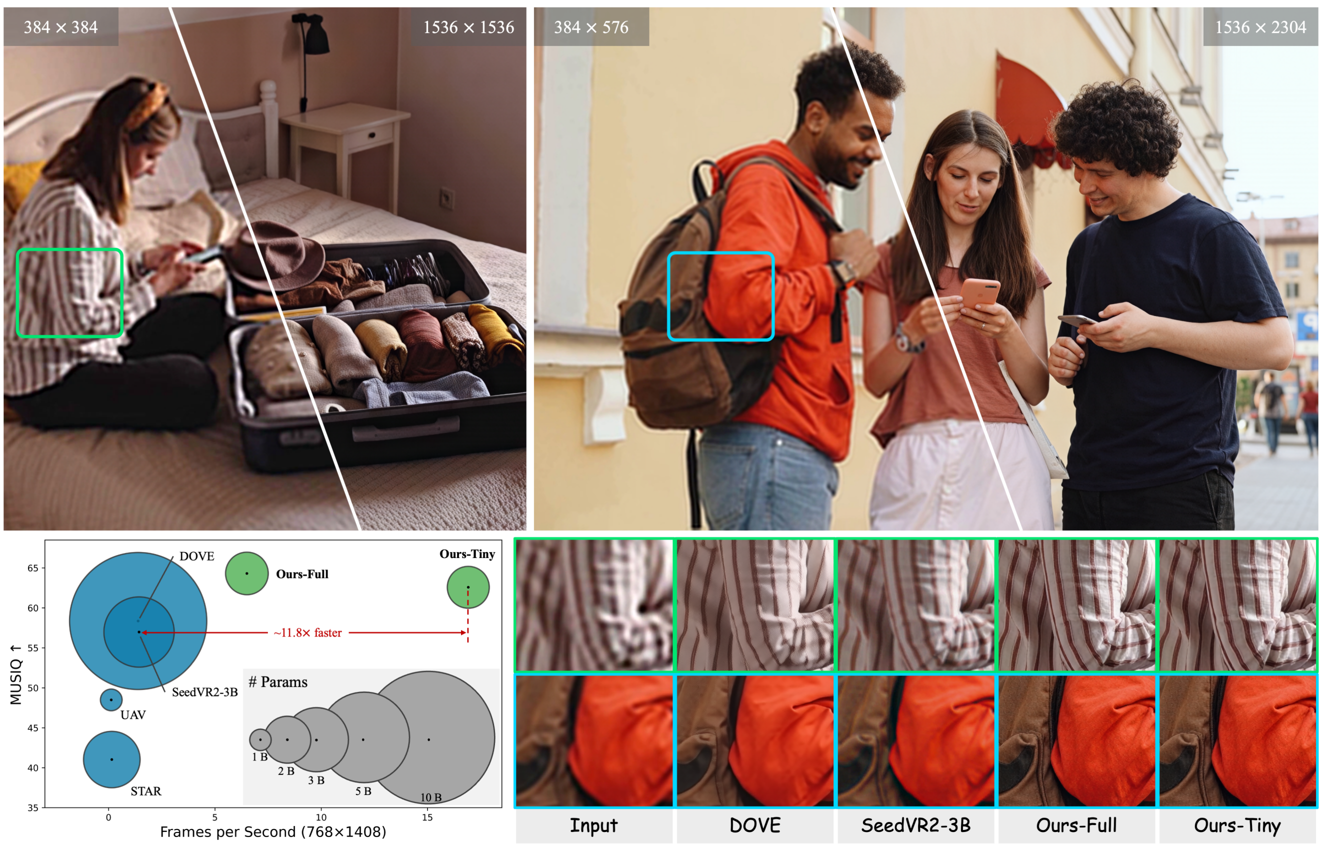

Figure 1. Efficiency and performance comparison in high-resolution video restoration. Compared to state-of-the-art VSR models (e.g., DOVE and SeedVR2-3B), FlashVSR restores sharper textures and more detailed structures overall. It achieves near real-time performance at 17 FPS on 768 × 1408 videos using a single A100 GPU, corresponding to a measured 11.8× speedup over the fastest one-step diffusion VSR model. **(Zoom in for best view)**.

Abstract

*Diffusion models have recently advanced video restoration, but applying them to real-world and AIGC-generated video super-resolution (VSR) remains challenging due to high latency, prohibitive computation, and poor generalization to ultra-high resolutions. Our goal in this work is to make diffusion-based VSR practical by achieving efficiency, scalability, and near real-time performance. To this end, we propose **FlashVSR**, the first diffusion-based one-step streaming framework for efficient video super-resolution. FlashVSR runs at ∼17 FPS for* 768 × 1408 *videos on a single A100 GPU by combining three complementary innovations: (i) a train-friendly three-stage distillation pipeline that enables streaming super-resolution, (ii) locality-constrained sparse attention that reduces redundant computation while bridging the train–test resolution gap, and (iii) a tiny conditional decoder that accelerates reconstruction without sacrificing quality. To support large-scale training, we also construct **VSR-120K**, a new dataset containing 120 K videos and 180 K images. Extensive experiments demonstrate that FlashVSR scales reliably to ultra-high resolutions and achieves state-of-the-art performance with up to ∼12× speed-up over prior one-step diffusion-based VSR models. We will release the code, pretrained models, and dataset to foster future research in efficient diffusion-based video super-resolution.*

1. Introduction

Video super-resolution (VSR) aims to recover high-quality frames from degraded videos, and plays an essential role in mobile photography, live streaming, and AIGC content. Recent VSR methods [3–5, 20, 24, 42, 53, 57, 69] and diffusion-based restorers [10, 40, 41, 53] have improved quality remarkably, yet real-time, high-resolution, and infinitely long video restoration remains unsolved.

However, achieving high-resolution, high-quality, and real-time streamable video super-resolution remains highly challenging, particularly for diffusion-based VSR. We identify three major obstacles: **(1)** High lookahead latency from chunk-wise processing. Due to memory constraints, most methods split long videos into overlapping segments and process them independently, leading to redundant computation and substantial lookahead latency. **(2)** The high computational cost of dense 3D attention. Full spatiotemporal attention scales quadratically with resolution, making it prohibitively expensive for long or high-resolution videos. **(3)** A train–test resolution gap. Most attention-based VSR models trained on medium-resolution videos degrade notably at higher resolutions (e.g., 1440p). To compensate, prior methods rely on spatial tiling during inference, which again introduces redundant computation. Our analysis reveals that this gap mainly arises from mismatched positional encoding ranges between training and inference.

In this work, we handle all three of these challenges and build FlashVSR, the first diffusion-based video super-resolution model towards real-time and streamable processing. As shown in Fig. 1, FlashVSR achieves near real-time inference at 17 FPS at a resolution of 768×1408 on a single A100 GPU. This is significantly faster than all other diffusion-based VSR methods, including STAR [53] ($\sim 120\times$ faster) and the strongest one-step diffusion-based baseline SeedVR2-3B [40] ($\sim 12\times$ faster). Additionally, benefiting from its streaming design we will soon introduce, FlashVSR only introduces 8 frames of lookahead latency, whereas previous chunk-based methods introduce latency equal to the segment length (~ 80 frames). FlashVSR further scales reliably up to 1440p, delivering detail-rich videos, as shown in the right example of Fig. 1; more visual results are provided in the appendix. FlashVSR achieves these advances through three key innovations.

Training-friendly distillation for one-step streaming VSR. We propose a three-stage distillation pipeline that progressively transforms a full-attention teacher into a one-step student with block-sparse causal attention. This design enables fully parallel frame training. In contrast, previous autoregressive video diffusion models [16, 22] require sequential frame generation during training, leading to strong serial dependencies, lower throughput, and much slower optimization, thus limiting scalability.

Sparse attention with locality constraints. To reduce the redundancy of dense 3D self-attention in VSR, we adopt block-sparse attention for more efficient spatiotemporal modeling. We first compute a coarse region-level attention map by pooling key–value features and then apply full attention only to the top-k most relevant regions. For high-resolution inputs, we further impose spatial local windows to restrict each query's attention range, aligning the effective positional encoding ranges between training and inference. These designs substantially improve high-resolution generalization and enable inference *without spatial tiling*. To our knowledge, this is the first diffusion-based VSR model employing sparse attention.

Tiny conditional decoder. After the accelerating diffusion transformer block (DiT), the causal 3D VAE decoder becomes the primary runtime bottleneck, consuming nearly 70% of the inference time at 768×1408 resolution. To address this, we introduce a tiny conditional decoder that leverages LR frames as auxiliary inputs in addition to latents, thereby simplifying HR reconstruction and enabling a more compact design. With the same parameter budget, our ablation shows that it outperforms its unconditional variant, while preserving visual quality comparable to the original VAE decoder and reducing decoding time to roughly $1/7$.

Additionally, we construct a new large-scale dataset, **VSR-120K**, comprising 120k videos (average length $>$350 frames) and 180k high-quality images, filtered via automated quality control. Existing VSR datasets are limited in scale and video quality, *e.g.*, DOVE contains only 2K videos. Our dataset enables joint image-video training and will be publicly released to advance VSR research.

Extensive quantitative and qualitative results show that FlashVSR achieves state-of-the-art performance with up to $\sim 12\times$ speedup, while locality-constrained attention mitigates the resolution-induced train–test gap for robust generalization to ultra-high-resolution videos.

2. Related work

2.1. Real-world video super-resolution

Early research in video super-resolution (VSR) often relied on simple synthetic degradations (*e.g.*, bicubic downsampling) [3–5, 13, 19, 24, 42], but these methods perform poorly on real-world data. To mitigate this gap, some works collected paired data from consumer devices [47, 56], but these datasets are small and sensor-biased. Later approaches introduced composite degradations combining blur, noise, and compression artifacts [6, 43, 53, 56, 57, 69].

The recent success of diffusion models in image and video generation and restoration [1, 12, 31, 48, 62, 63, 66, 70–72] has motivated their use in VSR [20, 53, 57, 67, 69]. For instance, Upscale-A-Video [69] employs optical-flow-guided propagation, MGLD-VSR [57] introduces motion-aware objectives, and DiffVSR [20] adopts staged optimization. More recent frameworks leverage video diffusion priors [2, 38, 58], such as SeedVR [40, 41], and DOVE [10]. While effective in enforcing temporal consistency, these methods are hindered by the high cost of 3D attention and reliance on chunk-based inference, leading to large lookahead latency and inefficiency, even in one-step models.

2.2. Streaming Video Diffusion Models

Practical VSR must handle sequences lasting minutes or longer, making streaming capability essential for deployment. Recent studies have extended diffusion models from short clips to long streaming video generation. Diffusion Forcing [7] reformulates denoising as block-wise sequential processing, enabling causal decoding. Follow-up works [8, 14, 16, 22, 37, 61] combined causal attention, KV-cache, and distillation to compress multi-step diffusion into a few denoising steps, enabling efficient online generation. However, these works primarily focus on video synthesis rather than restoration tasks such as VSR. Moreover, recent autoregressive video diffusion models rely on "student forcing" training to mitigate error accumulation, which requires predicting the previous output. This leads to serial unfolding of the forward process, reducing training efficiency.

2.3. Video Diffusion Acceleration

Despite their strong performance, diffusion models remain computationally expensive, limiting real-time applications. Existing acceleration strategies mainly fall into three categories: **Feature caching.** Works such as DeepCache [54], FasterDiffusion [18], and recent adaptations for video diffusion transformers(DiT) [9, 25, 30, 33] reduce redundancy by reusing intermediate activations. **One-step distillation.** Methods based on rectified flows [26, 27], score distillation [21, 68], and adversarial training [46, 60] compress iterative denoising into a single step, with representative examples including OSEDiff [51], SinSR [44], and TSD-SR [12]. Extensions to VSR such as DOVE [10] and SeedVR2 [40] achieve competitive results. **Sparse attention.** FlashAttention [11] improves memory and computation efficiency while also supporting block-sparse mechanisms, thereby reducing memory usage and accelerating attention computation. Techniques such as Sparse VideoGen [52], Sparse VideoGen2 [55], and others [34, 64] exploit spatiotemporal or semantic sparsity to reduce memory and computation while maintaining quality.

Despite recent advances, most methods remain inefficient, high-latency, and poorly generalized at high resolutions. In contrast, we introduce FlashVSR, the first diffusion-based VSR unifying one-step distillation, a train-friendly streaming design, and locality-constrained sparse attention. With a tiny conditional decoder for video restoration, FlashVSR addresses the key challenges of efficiency, temporal scalability, and high-resolution generalization, achieving near real-time performance and moving diffusion-based VSR closer to practical deployment.

3. Method

We present FlashVSR, an efficient diffusion-based one-step streaming framework for video super-resolution (VSR), achieving near real-time inference (17 FPS at 768×1408 on a single A100 GPU). Furthermore, to train a high-quality VSR, we also construct a large-scale high-quality dataset, VSR-120K. As illustrated in Fig. 2, FlashVSR builds on a three-stage distillation framework, enhanced with locality-constrained sparse attention to mitigate the train–infer resolution gap, and a tiny conditional decoder that reduces the cost of the 3D VAE decoder. Details are introduced below.

3.1. VSR-120K Dataset

To overcome the limited scale and quality of existing VSR datasets, we construct **VSR-120K**, a large-scale dataset for joint image–video super-resolution training. We collect raw data from open repositories such as Videvo, Pexels, and Pixabay[1], containing 600k video clips and 220k high-resolution images. For quality control, we employ LAION-Aesthetic predictor [32] and MUSIQ [17] for visual quality, and RAFT [36] for motion filtering. The final dataset consists of 120k videos (average length >350 frames) and 180k high-quality images. We will release VSR-120K for public research use. See Appendix for details.

3.2. Three-Stage Distillation Pipeline

To build a high-quality and efficient VSR model, we design a three-stage distillation pipeline: (1) joint video–image training to establish a strong teacher, (2) causal sparse attention adaptation for streaming efficiency, and (3) distribution-matching distillation into a one-step student.

[1] https://www.videvo.net/, https://www.pexels.com/, https://pixabay.com/

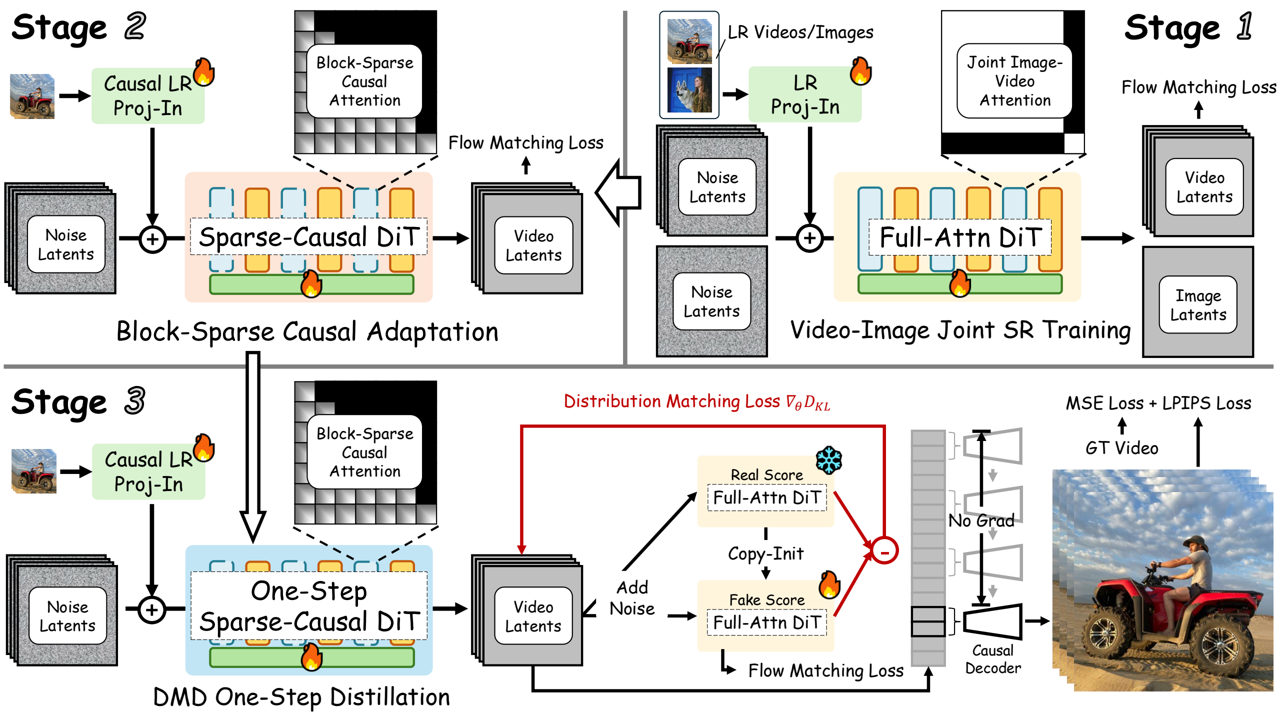

Figure 2. Overview of the three-stage training pipeline of FlashVSR, covering video–image joint SR training, adaptation with block-sparse causal attention for streaming inference, and distribution-matching one-step distillation combined with reconstruction supervision.

Stage 1. Video–Image Joint SR Training. We adapt a pre-trained video diffusion model (WAN2.1 1.3B [38]) to super-resolution by jointly training on videos and images, treating images as single-frame videos (f=1) to enable a unified 3D attention formulation. As illustrated in Stage 1 of Fig. 2, we apply a block-diagonal segment mask that restricts attention within the same segment:

$$\alpha_{ij} = \frac{\exp\left(\frac{q_i k_j^\top}{\sqrt{d}}\right) \mathbf{1}[\,\mathrm{seg}(i) = \mathrm{seg}(j)\,]}{\sum_l \exp\left(\frac{q_i k_l^\top}{\sqrt{d}}\right) \mathbf{1}[\,\mathrm{seg}(i) = \mathrm{seg}(l)\,]}, \tag{1}$$

where $\mathrm{seg}(i)$ denotes the segment identity of token i (image or video clip) and α denotes the normalized attention weight. Block-sparse constraints are omitted so the teacher retains full spatiotemporal priors. A fixed text prompt is used for conditioning, with cross-attention keys and values reused across samples. We further introduce a lightweight low-resolution(LR) Proj-In layer to project LR inputs into the feature space instead of using the VAE encoder. Training employs the standard flow matching loss [23].

Stage 2. Block-Sparse Causal Attention Adaptation. We adapt the Stage 1 full-attention DiT into a Sparse-Causal DiT by introducing causal masking and block-sparse attention, as illustrated in Fig. 2. The causal mask restricts each latent to its current and past positions. Following [34, 64], queries and keys are partitioned into non-overlapping blocks of size $(2, 8, 8)$ and reshaped into $(B, block\_num, 128, C)$ with $block\_num = L/128$. Within each block, average pooling yields compact block-level features to compute a coarse block-to-block attention map. The top-k most relevant block pairs are selected, and full 128×128 attention is applied only to these regions with the original (Q, K, V). This reduces attention cost to 10–20% of the dense baseline without performance loss. The LR Proj-In layer is converted to a causal variant for streaming inference, and training continues with the flow matching loss on video data only.

Stage 3. Distribution-Matching One-Step Distillation. Recent studies on one-step streaming video diffusion focus on video generation, typically requiring clean past frames as input for motion plausibility. Teacher forcing (conditioning on ground truth) [22] causes error accumulation during inference, while student forcing (conditioning on predicted latents) [16, 22] alleviates this but requires sequential unfolding, reducing efficiency.

In Stage 3 (Fig. 2), we refine the Stage 2 sparse-causal DiT into a one-step model G_{one} and propose a parallel training paradigm for streaming VSR. The model takes LR frames and Gaussian noise as input, with all latents trained under a unified timestep using a block-sparse causal attention mask. The Stage 1 full-attention DiT serves as the teacher G_{real}, while its copy G_{fake} learns the distribution of fake latents, following the DMD pipeline by Yin et al. [60]. Here, z_{pred} denotes the predicted latent, and x_{pred} the reconstructed HR frame. Due to memory constraints, two latents are randomly selected per iteration for decoding, with previous ones detached from gradients. The overall objective combines distribution-matching distillation, flow matching,

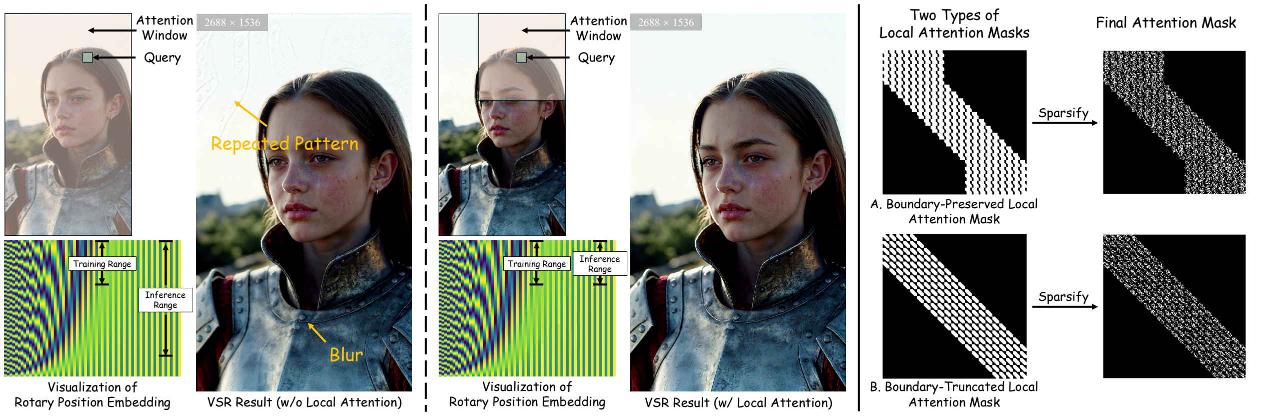

Figure 3. Locality-Constrained Sparse Attention. Left: At ultra-high resolutions, performing inference beyond the trained positional encoding range produces artifacts (e.g., repetition or blur). Restricting each query to a local attention window keeps the positional encoding range consistent between training and inference, thereby preventing artifacts. Right: Two local window rules, namely boundary-preserved and boundary-truncated, are illustrated. The final sparse attention mask is computed within these local masks.

and pixel-space reconstruction losses, with $\lambda = 2$:

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{DMD}}(z_{\text{pred}}, G_{\text{one}}, G_{\text{real}}, G_{\text{fake}})}_{\text{distribution-matching distillation}} + \underbrace{\mathcal{L}_{\text{FM}}(z_{\text{pred}}, G_{\text{fake}})}_{\text{flow matching}}$$
$$+ \underbrace{\|x_{\text{pred}} - x_{\text{gt}}\|_2^2 + \lambda\, \mathcal{L}_{\text{lpips}}(x_{\text{pred}}, x_{\text{gt}})}_{\text{decoder reconstruction}}. \tag{2}$$

Since training and inference rely only on LR frames and noise, the train–infer gap is eliminated. As a one-step model, the later layers of G_{one} already contain clean latent information propagated via the KV-cache for temporal continuity. *The core insight is that*, unlike video generation, VSR is strongly conditioned on LR frames, so clean historical latents are unnecessary for motion plausibility. The model focuses on content reconstruction, while temporal consistency is refined in later layers via the KV-cache. This design enables efficient parallel training while preserving high fidelity and eliminating the train–infer gap. Further discussion is provided in Appendix.

3.3. Locality-Constrained Sparse Attention

We observed that for super-resolution, models trained on medium-resolution data may fail to generalize to ultra-high resolutions (e.g., 1440p), leading to repeated patterns and blurring, as illustrated in Fig. 3. Our analysis shows that this arises from the periodicity of positional encodings: when the positional range at inference time far exceeds that seen during training, certain RoPE dimensions repeat their patterns, degrading self-attention, as shown in the bottom of Fig. 3. Because of this effect, many existing VSR methods resort to *spatial tiling* at high resolutions to limit the attention range and avoid positional overflow, at the cost of redundant computation and potential boundary artifacts.

To address this, we introduce **locality-constrained attention**, which restricts each query at inference to a *limited spatial neighborhood*, thereby aligning the effective attention range with that used during training. With the relative formulation of RoPE [35], this simple constraint eliminates the train–inference mismatch in positional range. As shown in the middle of Fig. 3, this approach closes the resolution gap and enables consistent performance even on ultra-high-resolution outputs, *without requiring any spatial tiling*. This design is further validated through quantitative results in Sec. 4.4 and qualitative visualizations in Appendix.

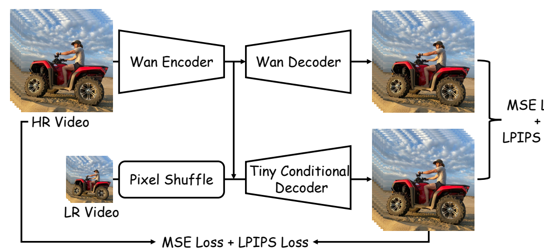

Figure 4. Training pipeline of the TC Decoder.

3.4. Tiny Conditional Decoder

We find that the VAE decoder dominates inference ($\sim 70\%$ runtime) and becomes the bottleneck. To address this, we design a **Tiny Conditional Decoder** (TC Decoder, Fig. 4) that conditions reconstruction on both LR frames and latents, rather than simply shrinking the original VAE decoder. This reduces decoding complexity and preserves fine details. The TC Decoder achieves nearly $7\times$ faster decoding than the original VAE decoder while maintaining comparable quality, and under the same parameter budget it consistently outperforms unconditional tiny decoders.

During training, we combine pixel-level supervision with distillation from the original Wan decoder. Let x_{pred} denote the reconstructed HR frame, x_{gt} the ground truth, and x_{wan} the Wan decoder output. The training objective is:

$$\begin{aligned}\mathcal{L} = &\|x_{\text{pred}} - x_{\text{gt}}\|_2^2 + \lambda\, \mathcal{L}_{\text{LPIPS}}(x_{\text{pred}}, x_{\text{gt}}) \\ &+ \|x_{\text{pred}} - x_{\text{wan}}\|_2^2 + \lambda\, \mathcal{L}_{\text{LPIPS}}(x_{\text{pred}}, x_{\text{wan}}).\end{aligned} \tag{3}$$

Table 1. Quantitative comparison on synthetic (YouHQ40, REDS, SPMCS), real-world (VideoLQ), and AIGC (AIGC30) datasets, covering diverse video domains. Best results are marked in **red** and second-best in **blue**.

Dataset	Metric	Upscale-A-Video	STAR	RealViformer	DOVE	SeedVR2-3B	Ours-Full	Ours-Tiny
YouHQ40	PSNR ↑	23.19	23.19	23.67	**24.39**	23.05	23.13	23.31
	SSIM ↑	0.6075	0.6388	0.6189	**0.6651**	0.6248	0.6004	0.6110
	LPIPS ↓	0.4585	0.4705	0.4476	0.4011	0.3876	0.3874	**0.3866**
	NIQE ↓	4.834	7.275	**3.360**	4.890	3.751	3.382	3.489
	MUSIQ ↑	43.07	35.05	62.73	61.60	62.31	**69.16**	66.63
	CLIPIQA ↑	0.3380	0.2974	0.4451	0.4437	0.4909	**0.5873**	0.5221
	DOVER ↑	6.889	7.363	9.739	11.29	12.43	**12.71**	12.66
REDS	PSNR ↑	24.84	24.01	**25.96**	25.60	24.83	23.92	24.11
	SSIM ↑	0.6437	0.6765	0.7092	**0.7257**	0.7042	0.6491	0.6511
	LPIPS ↓	0.4168	0.371	**0.2997**	0.3077	0.3124	0.3439	0.3432
	NIQE ↓	3.104	4.776	2.722	3.564	3.066	**2.425**	2.680
	MUSIQ ↑	53.00	46.25	63.23	65.51	61.83	**68.97**	67.43
	CLIPIQA ↑	0.2998	0.2807	0.3583	0.4160	0.3695	**0.4661**	0.4215
	DOVER ↑	6.366	6.309	8.338	**9.368**	8.725	8.734	8.665
SPMCS	PSNR ↑	23.95	23.68	**25.61**	25.46	23.62	23.84	24.02
	SSIM ↑	0.6209	0.6700	0.7030	**0.7201**	0.6632	0.6346	0.6450
	LPIPS ↓	0.4277	0.3910	0.3437	**0.3289**	0.3417	0.3436	0.3451
	NIQE ↓	3.818	7.049	3.369	4.168	3.425	**3.151**	3.302
	MUSIQ ↑	54.33	45.03	65.32	69.08	66.87	**71.05**	69.77
	CLIPIQA ↑	0.4060	0.3779	0.4150	0.5125	0.5307	**0.5792**	0.5238
	DOVER ↑	5.850	4.589	8.083	**9.525**	8.856	9.456	9.426
VideoLQ	NIQE ↓	4.889	5.534	**3.428**	5.292	5.205	3.803	4.070
	MUSIQ ↑	44.19	40.19	**57.60**	45.05	43.39	55.48	52.27
	CLIPIQA ↑	0.2491	0.2786	0.3183	0.2906	0.2593	**0.4184**	0.3601
	DOVER ↑	5.912	5.889	6.591	6.786	6.040	**8.149**	7.481
AIGC30	NIQE ↓	5.563	6.212	4.189	4.862	4.271	**3.871**	4.039
	MUSIQ ↑	47.87	38.62	50.74	50.59	50.53	**56.89**	55.80
	CLIPIQA ↑	0.4317	0.3593	0.4510	0.4665	0.4767	**0.5543**	0.5087
	DOVER ↑	10.24	11.00	11.24	12.34	12.48	**12.65**	12.50

Table 2. Efficiency comparison (peak memory, runtime, and parameters) on 101 × 768 × 1408 videos.

Metric	Upscale-A-Video	STAR	DOVE	SeedVR2-3B	Ours-Full	Ours-Tiny
Peak Mem. (GB)	18.39	24.86	25.44	52.88	18.33	11.13
Runtime (s) / FPS	811.71 / 0.12	682.48 / 0.15	72.76 / 1.39	70.58 / 1.43	15.50 / 6.52	5.97 / 16.92
Params (M)	1086.75	2492.90	10548.57	3391.48	1780.14	1752.18

4. Experiments

4.1. Implementation Details

FlashVSR is built upon Wan 2.1–1.3B [38] and fine-tuned with LoRA [15] (rank 384). All stages are trained on the VSR-120K dataset, with paired LR–HR videos and images synthesized via the RealBasicVSR degradation pipeline [6]. Training is conducted on 32 A100-80G GPUs, while all experiments are evaluated on a single A100 for consistency. All stages use a batch size of 32, taking about 2, 1, and 2 days for Stages 1–3, respectively. Stage 1 uses 89-frame clips (768 × 1280) and paired images; Stage 2 continues training with videos only; Stage 3 adopts the same setting. The AdamW optimizer [28] is used with learning rate 1×10^{-5} and weight decay 0.01. The TC Decoder is trained separately on 61-frame clips (384 × 384) for about 2 days.

4.2. Datasets, Metrics, and Baselines

We evaluate on three synthetic datasets (YouHQ40 [69], REDS [29], SPMCS [59]), one real-world dataset (VideoLQ [6]), and an AI-generated set (AIGC30). Synthetic LR frames are generated using the same degrada-

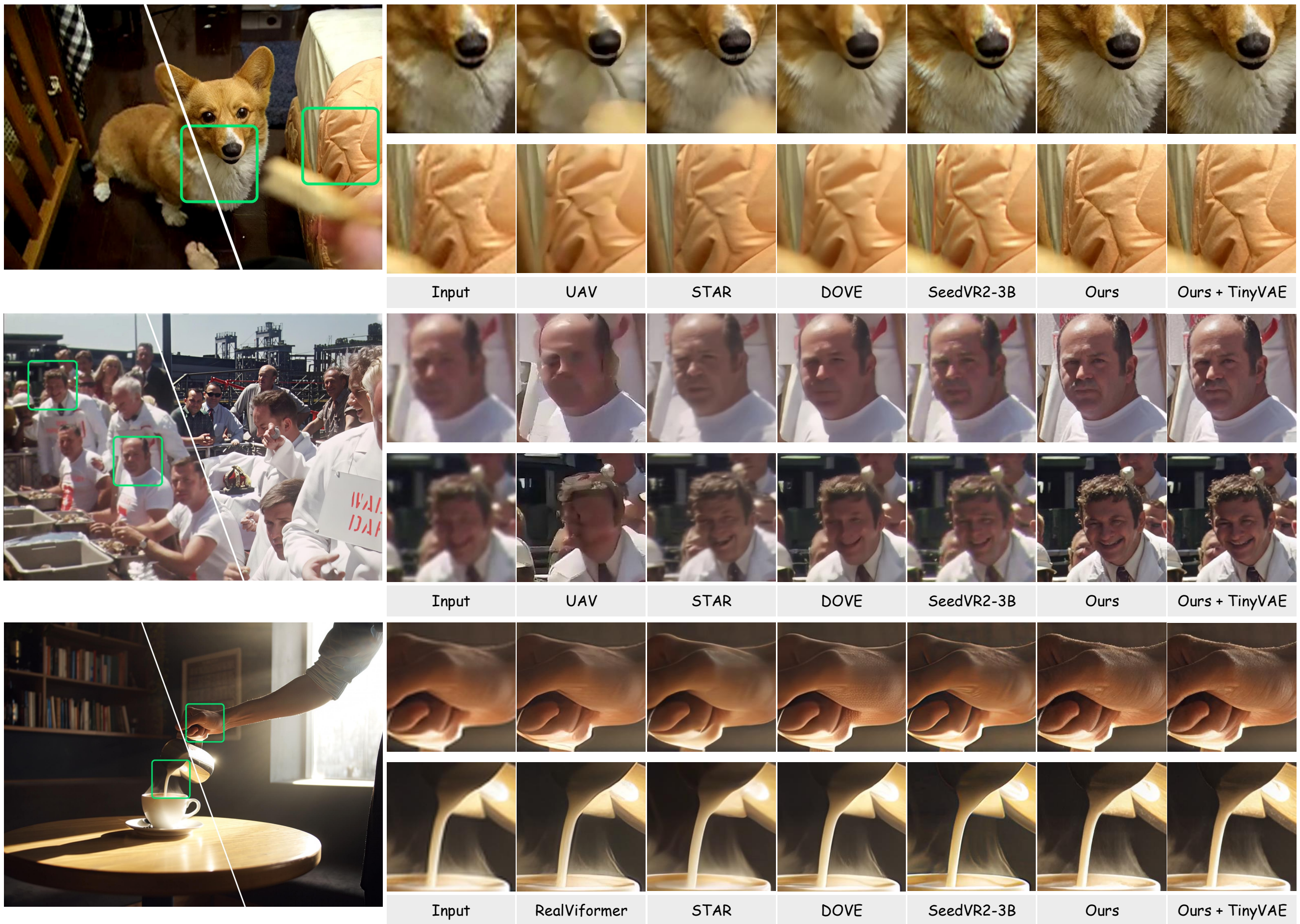

Figure 5. Visualization results of video super-resolution on real-world and AIGC videos.

Table 3. 13.6% sparse attention achieves similar reconstruction quality to full attention with ∼7.4× lower computation.

Method	PSNR ↑	SSIM ↑	LPIPS ↓	NIQE ↓	MUSIQ ↑	CLIPIQA ↑	DOVER ↑
13.6% Sparse	24.11	0.6511	0.3432	2.680	67.43	0.4215	8.665
Full Attention	24.65	0.6630	0.3320	2.878	65.77	0.4110	8.750

Table 4. Ablation of tiny conditional decoder.

Metric	Wan Decoder	Ours	Unconditional Variant
PSNR ↑	32.58	31.08	29.96
SSIM ↑	0.9417	0.9244	0.9079
LPIPS ↓	0.0715	0.1014	0.1231

tion pipeline as training. We evaluate with PSNR [49], SSIM [45], LPIPS [65], MUSIQ [17], CLIPIQA [39], and DOVER [50] on datasets with ground truth (YouHQ40, REDS, SPMCS), and only with the no-reference metrics (MUSIQ, CLIPIQA, DOVER) on datasets without ground truth (VideoLQ, AIGC30). We compare FlashVSR against RealViFormer [67] (non-diffusion transformer), STAR [53] and Upscale-A-Video [69] (multi-step diffusion), and DOVE [10] and SeedVR2-3B [40] (one-step diffusion). We also report results using the Wan Decoder (Ours-Full) and the proposed TC Decoder (Ours-Tiny).

4.3. Comparison with Existing Methods

Quantitative Comparisons. We compare FlashVSR with state-of-the-art real-world VSR methods, using default sampling settings for multi-step diffusion models (15 steps for STAR and 30 for Upscale-A-Video). As shown in Tab. 1, FlashVSR outperforms all competitors across datasets, with notable gains on perceptual metrics such as MUSIQ, CLIPIQA and DOVER. The proposed TC Decoder also improves reconstruction quality over the original Wan VAE decoder while remaining efficient. Although our reconstruction metrics are slightly lower, these metrics are known

Table 5. Ablation of Locality-constrained Attention. Both Boundary-Truncated and Boundary-Preserved outperform Global Attention across all metrics. Best in **red**, second-best in <u>blue</u>.

Method	PSNR ↑	SSIM ↑	LPIPS ↓	NIQE ↓	MUSIQ ↑	CLIPIQA ↑	DOVER ↑
Global Attention	24.21	0.6988	0.3423	3.152	65.57	0.5594	9.1259
Boundary-Truncated	<u>24.60</u>	<u>0.7155</u>	<u>0.3385</u>	**2.850**	**67.47**	**0.6278**	**9.5132**
Boundary-Preserved	**24.87**	**0.7232**	**0.3304**	<u>2.968</u>	<u>67.16</u>	<u>0.6029</u>	<u>9.3259</u>

to favor smoother outputs; visual results clearly indicate that FlashVSR restores sharper and more faithful details. Overall, the results demonstrate the effectiveness of FlashVSR in high-quality video super-resolution.

Qualitative Comparisons. To provide a more intuitive comparison of visual quality in real-world scenarios, we present qualitative results on VideoLQ and AIGC30 in Fig. 5. For clarity, we also enlarge selected patches to better illustrate the differences among the LR frames and outputs of all methods. FlashVSR produces sharper and more detailed reconstructions compared to baselines, with textures and structures that appear more natural. For instance, in the last row of Fig. 5, FlashVSR restores clearer hand textures and bookshelf details, yielding results that are visually more realistic. These qualitative observations are consistent with the quantitative improvements on perceptual metrics. Additional visualizations are provided in the Appendix.

Efficiency Analysis. Tab. 2 reports the efficiency comparison on 101-frame videos at 768×1408 resolution. With streaming inference, block-sparse attention, one-step distillation, and a lightweight conditional decoder, FlashVSR achieves substantial efficiency gains over all baselines. It runs $136\times$ faster than Upscale-A-Video (30 steps), $114\times$ faster than STAR (15 steps), and still $11.8\times$ faster than the fastest one-step model SeedVR2-3B, while also using much less peak memory (11.1 GB vs. 52.9 GB). STAR uses chunk-wise inference (chunk size 32, overlap 0.5), and most methods process entire sequences in one pass. In contrast, FlashVSR adopts streaming inference, reducing lookahead latency to just 8 frames (vs. 32 for STAR and 101 for others). These results demonstrate the practicality of FlashVSR for real-world deployment.

4.4. Ablation Study

Sparse Attention. We evaluate the impact of sparse attention on REDS. As shown in Tab. 3, FlashVSR with 13.6% sparsity achieves nearly identical reconstruction and perceptual quality compared to a full-attention baseline (KV-cache size 85 frames). At 768×1408, it reduces inference time from 1.105s to 0.355s per 8 frames ($3.1\times$ speedup), thereby substantially improving efficiency without compromising visual quality. This indicates that sparse attention effectively prunes redundant interactions, mitigating computational overhead while preserving essential spatiotemporal dependencies for high-quality video super-resolution.

Tiny Conditional Decoder. We evaluate the proposed TC Decoder on 200 unseen videos, where all inputs are encoded by the Wan VAE and reconstructed using three decoders: the original Wan decoder, our TC Decoder, and an unconditional variant. As shown in Tab. 4 and Fig. 5, the TC Decoder delivers visual quality nearly indistinguishable from the Wan decoder, with similarly close quantitative results. For a 101-frame 768×1408 video, it decodes in 1.60s versus 11.13s for the Wan decoder, achieving a $\sim 7\times$ speedup. It also consistently outperforms the unconditional variant across standard reconstruction metrics, confirming the clear benefit of conditioning on LR frames. Overall, the TC Decoder provides substantial acceleration with negligible fidelity loss, making it highly practical for real-world video super-resolution deployment.

Locality-constrained Attention. Fig. 3 illustrates that the proposed locality-constrained attention mask alleviates repeated patterns and blurring at ultra-high resolutions by aligning positional encoding ranges between training and inference. To quantitatively validate its effectiveness, we evaluate on 15 high-resolution videos (1536×2688, average 305 frames). We consider two variants depending on boundary handling (Fig. 3): Boundary-Preserved and Boundary-Truncated, both with a receptive field restricted to 1152×1152 and matched sparsity to global attention. Results are reported in Tab. 5. Compared to global attention, both variants consistently improve across all metrics. Notably, Boundary-Truncated achieves slightly higher perceptual quality, while Boundary-Preserved maintains competitive performance with better fidelity. These results confirm that locality-constrained attention effectively enhances VSR performance on ultra-high-resolution videos.

5. Conclusion

We presented FlashVSR, an efficient diffusion-based one-step framework for streaming video super-resolution. By combining streaming distillation, locality-constrained sparse attention, and a tiny conditional decoder, FlashVSR achieves state-of-the-art quality with near real-time efficiency and strong scalability to ultra-high resolutions. Our results demonstrate both its effectiveness and practicality, underscoring the significant potential of FlashVSR for real-world and AIGC-generated video applications.

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